

# Math 403 Homework #1, Spring 2025

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Collaborators: ...list those with whom you worked on this assignment...  
(1 point for each of up to 3 collaborators who also list you)

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Due: 11:59pm Saturday 25 January 2025

READING ASSIGNMENTS (item numbers are lecture numbers as in PDF lecture notes)

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1. for Thu. 9 January
  - [Hefferon, p.153–154] field axioms
  - [Cornell, “Fields” (about 1/3 of the way through 4330-week1.pdf)]: more on fields
  - [Climenhaga, §7.2, §9.1] isomorphism, rank-nullity
2. for Tue. 14 January
  - [Lax, Chapter 1] quotients
  - [Climenhaga, §5.1] quotients
  - [Cornell, “Quotient Spaces” (4330-week5.pdf)] universal properties
  - [Cornell, “Exact Sequences” (about halfway through 4330-week5.pdf)]
3. for Thu. 16 January
  - [Lax, Chapter 2] duality
  - [Climenhaga, §5.1, §8.2] duality, transpose
  - [Treil, §5.1–5.5] Hermitian inner products, adjoints; §5.2–5.4 should be mostly review
4. for Tue. 21 January
  - [Lax, Appendix 15] Jordan canonical form (very short proof)
  - [Hefferon, §5.IV.1] characteristic polynomial and minimal polynomial
  - [Treil, §9.3 and §9.5] generalized eigenspaces and Jordan canonical form
5. for Thu. 23 January
  - [Lax, Chapter 14, p.214–221] norms, equivalence, continuity, local compactness
  - [Stewart–Sun, §II.1] norms, equivalence, Hahn–Banach theorem
6. for Tue. 28 January
  - [Lax, Chap.12, p.187–190] Convex sets
7. for Thu. 30 January
  - [Serge Lang, *Linear Algebra*, Chapter XII, §1–§2] separating hyperplanes
  - [Serge Lang, *Linear Algebra*, Chapter XII, §3–§4] support hyperplane, extreme point

## EXERCISES

1. (Freshman's Dream): The *characteristic* of a field  $F$  is the smallest positive integer  $p$  such that the sum  $1 + \cdots + 1$  of  $p$  multiplicative identities is 0 in  $F$ . Prove that  $p$  is prime if it is finite. If  $F$  has characteristic  $p$ , show that  $(a + b)^p = a^p + b^p$  for  $a, b \in F$ .

Solution

If  $p=2$ , then you get the field with the elements 0 and 1. 2 is a prime and you can't have a field with less elements, otherwise the 0 and 1 elements wouldn't be distinct.

For the sake of notational ease take the symbol 2 to mean  $1+1$ .

$$2 := 1 + 1$$

$$3 := 1 + 1 + 1$$

$$4 := 1 + 1 + 1 + 1$$

$$\dots$$

$$p := 0 = 1 + 1 + 1 \dots$$

We first see that by using the axioms of a field, multiplication with this notation is like modulo multiplication.

Armed with our notation, let's attack the question.

Assume  $F$  is finite and its characteristic is a composite, meaning you can write  $p$  as multiple of two elements in the field  $a$  and  $b$  with  $a < p$  and  $b < p$ .

But this also implies that 0 would have an inverse because  $a * b * b^{-1} = a = 0 * b^{-1} = 0$ , so this implies  $0 = b^{-1}$ . But this is obviously nonsense.

So if  $F$  is finite  $p$  has to be prime.

Going to the second question, we can use the binomial theorem to get that

$$(a + b)^p = \sum_{k=0}^p \binom{p}{k} a^k b^{p-k} \tag{1}$$

If we look at the terms with  $k \neq 0$  and  $k \neq p$ , we get that  $\binom{p}{k} = p * ((p-1)! / ((p-k)!k!))$ , since  $p=0$ , then all these terms evaluate to 0 giving  $(a + b)^p = a^p + b^p$

2. Fix a vector space  $V$  and a subspace  $W \subseteq V$  over a field  $F$ . Let  $\pi : V \rightarrow V/W$  be the *projection homomorphism* given by  $\pi(v) = v + W$ . Write  $X$  for the set of all subspaces of  $V$  that contain  $W$ , and write  $Y$  for the set of all subspaces of  $V/W$ . Prove that  $\pi$  induces a bijection between these two sets, with

$$\begin{aligned} X &\rightarrow Y \\ L &\mapsto \pi(L) = \{\pi(v) \mid v \in L\} \end{aligned}$$

and

$$\begin{aligned} Y &\rightarrow X \\ M &\mapsto \pi^{-1}(M) = \{v \in V \mid \pi(v) \in M\}. \end{aligned}$$

3. Show that giving an exact sequence  $\cdots \rightarrow V_{i-1} \rightarrow V_i \rightarrow V_{i+1} \rightarrow \cdots$  is the same as giving a collection of short exact sequences  $0 \rightarrow K_i \rightarrow V_i \rightarrow K_{i+1} \rightarrow 0$ , one for each  $i$ . (The long exact sequence is said to be constructed by *splicing* the short exact sequences together.)
4. Rank-nullity theorem for exact sequences: Given an exact sequence

$$0 \rightarrow V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_r \rightarrow 0,$$

prove that  $\sum_{i=0}^r (-1)^i \dim V_i = 0$ .

5. Rank-nullity for arbitrary complexes: Given a complex  $0 \rightarrow V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_r \rightarrow 0$ , write  $B_i = \text{im}(V_{i-1} \rightarrow V_i)$  and  $Z_i = \ker(V_i \rightarrow V_{i+1})$ . Prove that

$$\sum_{i=0}^r (-1)^i \dim V_i = \sum_i (-1)^i \dim H_i,$$

where  $H_i = Z_i/B_i$  is the  $i^{\text{th}}$  homology of the complex. [In Exercise 4,  $B_i = Z_i = K_i$ , so  $H_i = Z_i/B_i = 0$  for all  $i$ .]

6. Two elements  $u$  and  $v$  in a vector space  $V$  are *congruent modulo* a subspace  $W \subseteq V$ , written  $u \equiv v \pmod{W}$ , if  $u + W = v + W$ . Show that congruence modulo  $W$  is an *equivalence relation* on  $V$ , meaning that it is
- reflexive:  $v \equiv v \pmod{W}$  for all  $v \in V$ ;
  - symmetric: if  $u \equiv v \pmod{W}$ , then  $v \equiv u \pmod{W}$ ; and
  - transitive: if  $u \equiv v \pmod{W}$  and  $v \equiv x \pmod{W}$  then  $u \equiv x \pmod{W}$ .
7. Give an example of three subspaces  $Y_1, Y_2$ , and  $Y_3$  in  $\mathbb{R}^2$  such that  $Y_1 + Y_2 + Y_3 = \mathbb{R}^2$  and  $Y_i \cap Y_j = \{\mathbf{0}\}$  for all  $i \neq j$ , but  $\mathbb{R}^2$  is not the direct sum of  $Y_1, Y_2$ , and  $Y_3$ .
8. Prove that if  $V$  is a vector space and  $W \subseteq V$  is a subspace, then  $W$  has a *complement*: a subspace  $U \subseteq V$  such that  $V = W \oplus U$ . Hint:  $V/W$  has a basis; lift it back to  $V$ .

9. For vectors  $\mathbf{x} = (1, 2i, 1 + i)$  and  $\mathbf{y} = (i, 2 - i, 3)$ , compute
  - (a)  $\langle \mathbf{x}, \mathbf{y} \rangle$ ,  $\|\mathbf{x}\|^2$ ,  $\|\mathbf{y}\|^2$ , and  $\|\mathbf{y}\|$ ; /3
  - (b)  $\langle 3\mathbf{x}, 2i\mathbf{y} \rangle$  and  $\langle 2\mathbf{x}, i\mathbf{x} + 2\mathbf{y} \rangle$ ; /3
  - (c)  $\|\mathbf{x} + 2\mathbf{y}\|$ . [Use parts (a) and (b) for this.] /3
10. Prove that for vectors in an inner product space,  $\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\operatorname{Re}\langle \mathbf{x}, \mathbf{y} \rangle$ . /3
11. For any  $m \times n$  complex matrix  $A$ , prove that  $\ker(A^*A) = \ker(A)$ . /3
12. Prove that if  $P$  is self-adjoint (that is,  $P^* = P$ ) and idempotent (that is,  $P^2 = P$ ) then  $P$  is the matrix for an orthogonal projection. /3
13. If  $V$  is a vector space over  $\mathbb{C}$  of dimension  $n$ , then it is also a vector space over  $\mathbb{R}$  of dimension  $2n$ . (If this isn't clear to you, then write down a proof.) Given a Hermitian inner product  $\langle \mathbf{x}, \mathbf{y} \rangle$  on  $V$  as a complex vector space, show that the real part  $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{R}} = \operatorname{Re}(\langle \mathbf{x}, \mathbf{y} \rangle)$  is an inner product on  $V$  as a real vector space. /3
14. Find all possible Jordan forms of linear transformations with characteristic polynomial  $(t - 1)^2(t + 2)^2$ . /3
15. Find all possible Jordan forms of linear transformations with characteristic polynomial  $(t - 2)^3(t + 1)$  and minimal polynomial  $(t - 2)^2(t + 1)$ . /3
16. How many similarity classes are there for  $3 \times 3$  matrices whose only eigenvalues are  $-3$  and  $4$ ? /3
17. Prove or disprove: two  $n \times n$  matrices are similar if and only if they have the same characteristic polynomial and minimal polynomial. /3